

1. Executive Summary

		Traffic Summary		Count			
		Total Flows Observed		18460			
Total Threats Mitigated		34		Total Packets Analyzed		916188093	
High Priority		7		Malicious Packets		915416319	
Low Priority		27		True Negatives Passed		13835	
Manual Releases		0					
Manual Blocks		0					
				Attack Vector		Count	
				ICMP Flood		6	
				SYN Flood		12	
				UDP Flood		6	
				Uncertain		10	
ML Metrics		Value		Action		Count	
False Positives		0		Quarantined		10	
FP Rate		0.00%		Time Ban		10	
IF Precision		100.00%		Blackhole		0	
IF Recall		92.96%		Released		0	
IF F1-Score		96.35%					
IF Accuracy		98.28%					

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	92.96%	Share of actual attacks successfully flagged
F1-Score	96.35%	Balanced measure combining Precision and Recall
Accuracy	98.28%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.04%	Actual attacks that went undetected
True Positive Rate (TPR)	92.96%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4051	FN 307
Actual: Normal	FP 0	TN 13528

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	2600	0	0
Act: ICMP	0	360	0
Act: UDP	0	0	1091

2c. Response Latency

Metric	Value	Description
Detection Time	1266.2 ms	Average time taken to detect an attack
Mitigation Response Time	25.4 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.43%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	36.03%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	7.61%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	2	2	2026-09-09 21:35:57	2026-09-09 21:37:08

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-09 21:33:41	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:33:42	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:33:45	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:33:46	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:33:52	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-09 21:33:52	10.0.0.17	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-09 21:33:52	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-09 21:33:53	10.0.0.17	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-09 21:33:56	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-09 21:33:56	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-09 21:33:56	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-09 21:33:57	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-09 21:34:12	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:13	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:15	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:23	10.0.0.20	DDoS	UDP Flood	76.6%	High	Time Ban (1m)
2026-09-09 21:34:23	10.0.0.20	DDoS	UDP Flood	76.6%	High	Time Ban
2026-09-09 21:34:24	10.0.0.21	DDoS	UDP Flood	79.6%	High	Time Ban (1m)
2026-09-09 21:34:24	10.0.0.21	DDoS	UDP Flood	79.6%	Critical	Time Ban
2026-09-09 21:34:26	10.0.0.22	DDoS	UDP Flood	79.6%	High	Time Ban (1m)
2026-09-09 21:34:26	10.0.0.22	DDoS	UDP Flood	79.6%	High	Time Ban
2026-09-09 21:34:42	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:43	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:50	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-09 21:34:52	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (2m)
2026-09-09 21:34:53	10.0.0.17	DDoS	SYN Flood	92.9%	Critical	Time Ban (2m)
2026-09-09 21:34:53	10.0.0.23	DDoS	ICMP Flood	91.9%	High	Time Ban (1m)
2026-09-09 21:34:55	10.0.0.24	DDoS	ICMP Flood	91.6%	High	Time Ban (1m)
2026-09-09 21:34:55	10.0.0.24	DDoS	ICMP Flood	91.6%	Critical	Time Ban
2026-09-09 21:34:57	10.0.0.19	DDoS	SYN Flood	92.8%	Critical	Time Ban (2m)
2026-09-09 21:34:57	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban (2m)
2026-09-09 21:34:59	10.0.0.23	DDoS	ICMP Flood	91.9%	Critical	Time Ban
2026-09-09 21:35:00	10.0.0.25	DDoS	ICMP Flood	91.7%	Critical	Time Ban (1m)
2026-09-09 21:35:02	10.0.0.25	DDoS	ICMP Flood	91.7%	Critical	Time Ban

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6347	92.6%	Critical	Phase 2	2	1m 11s	2026-09-09 21:37:08	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6347	92.6%	Critical	Phase 2	1	10s	2026-09-09 21:36:07	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE